

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (IT)

Dec 2013

IT 701 Distributed Computing & Systems

Date: 30.12.2013, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) What is distributed computing? How does distributed computing differ from parallel computing? [05]
- (b) What are the primary features of distributed computing? [02]
- Q - 2 (a) List and explain the core standards of web service. [03]
- (b) Draw and explain the Data-flow control in an RPC application. [05]

OR

- (b) Draw and explain Data-flow control in a Java RMI application [05]
- (c) What is Data Grid? List and explain the core functional data requirements for Data Grid. [06]

OR

- (c) What are the requirements of resources categories in grid computing? [06]
- Q - 3 (a) What is virtual organization? List the capabilities of virtual organization should have. [07]
- (b) Providing solution for distributed computing is challenging. List and explain the challenges in implementation of distributed computing. [07]

OR

- (b) What is Peer-to-Peer system? Explain the types if P2P system. How overlay network structure is used to design P2P system? [07]

SECTION – II

- Q - 4 (a) What is CORBA? Describe the need of IDL in CORBA. How does CORBA differ from RMI? [07]

Q - 5 (a) What is cluster computing? Explain how cluster computing differs from grid computing? [07]

(b) Write a java program to create client server socket program. Client should able to send a string and server will reverse the string send it back to server. (Use either TCP or UDP socket) [07]

OR

(b) What is DFS? Explain the following terms related to DFS: [07]

1. Naming and Location
2. Caching and Replication

Q - 6 (a) Write short note on following (ANY TWO): [14]

1. Chandy's Scheme for deadlock detection.
2. Reflective memory DSM Architecture.
3. Deadlock detection, Prevention and avoidance in distributed computing.
